PC 23CS202 Data Structures Lab 0 0 3 1.5

At the end of the course, students will be able to

Code	Course Outcomes	Mapping with POs / PSOs						
		PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
23CS202.1	Implement linear data structures in organizing and accessing data efficiently in algorithms	3	3	3	2	2	3	3
23CS202.2	Design, implement, and apply linked lists for dynamic data storage, demonstrating understanding of memory allocation	3	3	3	3	2	3	3
23CS202.3	Implement small scale programming challenges involving data structures such as stacks, trees	3	3	3	3	2	3	3
23CS202.4	Apply queue-based algorithms for efficient task scheduling and breadth-first traversal in graphs and distinguish between deques and priority queues, and apply them appropriately to solve data management challenges	3	3	3	3	2	3	3
23CS202.5	Recognize scenarios where hashing is advantageous, and design hash-based solutions for specific problems	3	3	3	3	2	3	3

List of Experiments

1. Write C programs to implement Array Manipulation — **COs: CO1**
2. Write C programs to implement operations on Linked List — **COs: CO2**
3. Write C programs to implement Linked List Applications — **COs: CO2**
4. Write C programs to implement operations on Double Linked List — **COs: CO2**
5. Write C programs to implement operations on Stack — **COs: CO3**
6. Write C programs to implement operations on Queue — **COs: CO4**
7. Write C programs to implement Stack Applications — **COs: CO3**
8. Write C programs to implement Queue Applications — **COs: CO4**
9. Write C programs to implement operations on Binary Search Tree — **COs: CO5**
10. Write C programs to implement Hashing — **COs: CO5**

Exercise Problems

1. Write a program to reverse an array
2. Write C programs to implement the searching techniques – linear & binary search
3. Write C programs to implement sorting techniques – bubble, selection and insertion sort
4. Implement a singly linked list and perform insertion and deletion operations
5. Develop a program to reverse a linked list iteratively and recursively
6. Solve problems involving linked list traversal and manipulation
7. Create a program to detect and remove duplicates from a linked list
8. Implement a linked list to represent polynomials and perform addition
9. Implement a double-ended queue (deque) with essential operations
10. Implement a doubly linked list and perform various operations to understand its properties and applications
11. Implement a circular linked list and perform insertion, deletion and traversal
12. Implement a stack using arrays and linked lists
13. Write a program to evaluate a postfix expression using a stack
14. Implement a program to check for balanced parentheses using a stack

15. Implement a queue using arrays and linked lists
16. Develop a program to simulate a simple printer queue system
17. Solve problems involving circular queues
18. Use a stack to evaluate an infix expression and convert it to postfix
19. Create a program to determine whether a given string is a palindrome or not
20. Implement a stack or queue to perform comparison and check for symmetry
21. Implement BST using linked list
22. Implement traversing of BST
23. Implement a hash table with collision resolution techniques
24. Write a program to implement a simple cache using hashing

References

1. Mark Allen Weiss, "Data Structures and Algorithm Analysis in C", 2nd Edition, Pearson, 2002
2. Ellis Horowitz, Sartaj Sahni, Susan Anderson-Freed, "Fundamentals of Data Structures in C", Silicon Press, 2008
3. Lab Manual for Data Structures, Department of Computer Science & Engineering, NSRIT

CONTROL COPY ATTESTED

Chairman
Board of Studies (CSE)